

Bachelor's Programme in Electrical and Electronics Engineering

Vibration Analysis of Faulty Electric Motor in the Frequency Domain

A Study on Spectral Centroid as Fault Indicator for Misalignment

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Abstract

Electric induction motors (IM) are widely used in various industries and are prone to faults such as shaft misalignment. Detecting these faults is important to minimize downtime and to save costs.

This study investigates the use of spectral centroid (SC) as a tool for detecting misalignment faults. Vibration data were collected and compared between a healthy IM, an IM with 0.5 mm of parallel misalignment, and an IM with 1 mm of parallel misalignment in all three axes of movement (X, Y, Z). A DC-generator was used as a load for the IM and data was collected under four different load conditions.

The results showed a trend of SC being larger for the vibration of the misaligned IM in the X and Y directions and under lighter loads in the Z direction. However, under higher loads, the differences in SC between healthy and misaligned IMs became smaller in all directions of data. In the Z direction, for larger loads, the SC for the healthy motor surpassed that of the misaligned IMs.

In addition, a statistical analysis was performed on the vibration signals, where measures such as skewness, kurtosis, mean, median, and standard deviation were calculated. This statistical analysis showed that the standard deviation differed between the motors. A 0.5 mm misalignment had a standard deviation that was between 10% and 20% greater than the intact motor, and a 1 mm misalignment had up to twice the standard deviation of the intact motor. Other statistical methods did not show large deviations between the intact and misaligned motors.

These findings suggest that SC has the potential to be an indicator of motors with misalignment faults, but in addition some limits, such as SC being orientation dependant. To create more robustness for the method, further research is necessary to validate the results, as the data set used in this study was limited.

Keywords spectral centroid, vibration analysis, induction motor, misalignment, fault detection

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Sammandrag

Elektriska asynkronmotorer (IM) används brett i olika industrier och är känsliga för fel som en felinriktad axel. För att minimera driftstopp och för att spara kostnader är det viktigt att kunna upptäcka denna typ av fel effektivt.

Denna studie undersöker användningen av spektral centroid (SC) som ett verktyg för att upptäcka felinriktningsfel. Data som användes i denna studie innehöll vibrationsdata från en hel asynkronmotor, en asynkronmotor med 0,5 mm parallell felinriktning och en asynkronmotor med 1 mm parallell felinriktning i alla tre axlar av rörelse (X, Y, Z). En DC-generator var använd för att skapa motstånd till asynkronmotorerna och simulera olika motståndstillstånd.

Resultaten från denna studie visade en tydlig trend där SC var högre för felinriktade motorerna i X- och Y-riktningen, samt för Z-riktningen under lägre motstånd. X-riktningen visade tydligaste skillnaderna, då hela motorn hade SC-värden mellan 45 Hz och 60 Hz, medan felinriktade motorerna hade värden mellan 59 Hz och 81 Hz. Med högre motstånd blev skillnaderna i SC mindre mellan hela och felinriktade motorerna i alla riktningar. För högre motstånd blev SC större för hela motorn än felinriktade motorerna i Z-riktningen.

Därtill gjordes en statistisk analys på vibrationssignalerna, där mått som skevhet, kurtosis (toppighet), medeltal, median och standardavvikelse mättes. Denna statistiska analys visade att standardavvikelsen skiljde sig mellan motorerna. 0,5 mm felinriktning hade en standardavvikelse som var mellan 10-20% större än hela motorn, och 1 mm felinriktning hade upp till dubbelt större standardavvikelse från hela motorn. Andra statistiska metoderna visade inte stora avvikelser mellan hela och felinriktade motorerna.

Dessa resultat indikerar att det finns potential för SC som en felindikator för motorer med parallell felinriktning, men också metodens begränsningar, som att den visade tendenser av att vara orientationsberoende. För att skapa mera pålitlighet i metoden krävs det mera forskning för att bekräfta de fynd som gjordes i denna studie, delvis på grund av den begränsade datamängden.

Nyckelord spektral centroid, vibrationsanalys, asynkronmotor, felinriktning, felidentifiering

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1 Introduction

1.1 Background to the study and research objective

Electric induction motors are widely used in various industries due to their efficiency, affordability, and low maintenance costs. With consistent use, such motors develop different faults over time, such as bearing wear, imbalances, or misalignments in shafts, which can cause permanent damage and increase costs through repairs and machine downtime.

Early detection of these faults is critical for avoiding unnecessary degradation or catastrophic sudden failures. Different methods have been studied for fault detection and fault diagnosis. Each method used in isolation has inherent limitations, such as reliability or sensitivity under certain conditions.

Motor vibration is a commonly used method for fault detection. It is cost effective and allows for remote monitoring once sensors have been installed on the machine.

This study aims to explore the use of spectral centroid as a tool for detecting the misalignment type of faults in induction motors. Studies such as [1] mention that misalignment is known to cause integer multiples in the frequency spectrum, which could cause a redistribution in vibrations towards higher frequencies. Spectral centroid, being a tool that calculates the centre of mass of a signal's frequency spectrum, could detect this kind of redistribution.

1.2 Research boundaries

The focus of this study is the vibration analysis of the misalignment fault in an electric induction motor. Although the database included other common faults, such as broken rotor bar or imbalance, they are outside the scope of this study. This study uses basic statistical measures, such as mean, median, kurtosis, skewness, and standard deviation, as well as Fourier analysis, to process and analyze the vibration signal. The in-depth analysis is done using the spectral centroid values for each signal. No advanced methods or machine learning were used for detecting differences in the signals.

For the misalignment fault, the database contains measurements for both direct-on-line starter (DOL) in varying load levels as well as variable frequency drive (VFD) data in different frequencies [2]. The database also included 0.5mm and 1mm misalignment in three axes of vibration. The number of recordings under each condition was limited to only one per condition [2]. This limits the statistical robustness of the study, and as a result, some conclusions were made based on trends found in the data. All available DOL data for the misalignment fault were used in this study. The DOL method was chosen for its consistent motor startup conditions. This avoids additional variables that could come up from varying speed and frequency controls.

2 Theoretical background

Signal processing is a broad and complex topic with a multitude of methods. This chapter presents the theoretical background for the specific methods used in this study and how they may be useful to us. The basic construction of an induction motor is also presented to give some context to the reader.

2.1 Construction of Induction motor

Three-phase induction motors mainly consist of two parts, the stator and the rotor. The stator is the stationary part that does not rotate and contains the outer frame, the stator winding, and the stator core [3].

There are two different types of rotors: a squirrel cage rotor and a phase wound rotor. The squirrel cage rotor has a laminated iron core of thin metal sheets with copper or aluminium bars that go from one end ring to the other, which are skewed for low power motors. The phase wound rotor also has a laminated core, and while the stator retains the same three phase windings as the squirrel cage motor, the rotor itself additionally contains windings. These windings are connected to external circuits via slip rings and brushes, which can wear out over time. [3]

Through the centre of each rotor passes the shaft, which transfers the mechanical power produced by the induction motor [3]. The misalignment in the shaft can be angular or parallel. Angular misalignment refers to when the shaft and load are at an angle to each other, forming a slight “V” shape. Parallel misalignment refers to when the centreline of the shaft and load may be parallel but are laterally offset [4]. This study explores data about parallel misalignment.

2.2 Signal processing and fault detection

An important tool used for signal processing in signal analysis and fault detection is the Fourier transformation. It is used to decompose the time domain of any given signal and give a visual presentation of its frequency spectrum. Fourier analysis and frequency spectrum evaluation have been proven to be useful tools for detecting faults in electric motors [5-7].

Studying the vibrations of a motor is not a particularly new concept. It has been found that restriction in the movements of the shafts by either housing or coupling causes harmonics of the rotor speed frequency, with a larger 2x component, as discussed in [1], which references Victor Wowk’s work on “Machinery Vibration” from 1991. The formula for calculating the rotor speed frequency, as mentioned in [1], is

$$f_r = (1 - s) \cdot \frac{f_s}{p} \quad (1)$$

where f_r is the rotor speed frequency, s is the rotor slip, f_s is the supply frequency, and p is the number of pole pairs. This will appear as a distinct peak when analysing the frequency spectrum of our signal.

To gauge the characteristics of a signal, general statistical measures like skewness, kurtosis, mean, median, and standard deviation were used in this study. The mean is useful for gauging if the DC signal was successfully removed to give a better

comparison between each signal. That is, to make sure that any baseline or offset shifts don't affect the analysis or results.

2.3 Spectral centroid

The spectral centroid (SC) is a frequency domain feature used to characterize the centre of mass of a signal's frequency spectrum. It has been shown to be useful in characterizing the condition of rotating machinery under different fault conditions [8], [9].

The SC is defined mathematically as

$$c = \frac{\sum f_k m_k}{\sum m_k} \quad (2)$$

here c represents the SC value, f_k represents frequency of bin k , and m_k represents the magnitude of the frequency at f_k . By calculating the SC values, we can observe shifts in terms of dominant frequencies of the signal.

3 Description of previous studies

Throughout the use of induction motors, different methods have been explored for fault detection. One of these methods is motor current signature analysis (MCSA). MCSA has been widely used as a way for fault detection in electric motors since the 1980s [10]. MCSA is a non-invasive analysis technique, which makes it a popular choice for condition monitoring due to its remote monitoring capabilities. MCSA has been shown to be effective at distinguishing certain types of faults. However, it may not reliably detect all fault types [11]. One study specifically noted that MCSA encounters difficulties in consistently detecting misalignment faults and recommended that supplementary tests be conducted [11].

Previous studies, such as [12], experimented with machine learning algorithms as a tool to get more precise results and to minimize false positives and negatives. In this study, the authors compared vibration and current-based fault detection with stray flux in parallel misalignment. The study found that vibration and current analysis were significantly more effective than stray flux, and combined with machine learning, they managed to get even more precise fault detection.

In [11], they compared motor current signature analysis (MCSA) directly to vibration analysis of squirrel cage induction motors for different malfunctions. The paper found that both methods were effective at detecting faults, but the results varied under different conditions. The vibration analysis gave more precise conclusions for unloaded conditions, while MCSA was more effective at fault detection when the motor was operating closer to full load conditions. Additionally, they found that the $3f_r$ and $4f_r$ components in the frequency spectrum were resilient to load changes.

The study in [9] proposed to use the spectral centroid indicator (SCI) in combination with minimum entropy convolution (MED) filtering for detecting lubrication-starved bearings. The problem they tried to tackle was that lubrication-starved bearings caused impulses at irregular and random intervals. This caused problems for detecting these faults in the frequency spectrum using typical fault detection methods for bearings, such as kurtosis and root-mean-square (RMS) values. The MED proved effective at filtering out the irregular impulses. Comparing the SCI between the normal and lubrication-starved bearing of their unfiltered signals, the difference was less than 20%. After MED filtering, the difference in SCI jumped to a factor of close to tenfold. This indicates that SCI can be useful but may be a limited tool on its own and can become significantly more effective with adequate signal processing.

4 Data analysis

This study uses data provided by the Electrical Energy Conversion System Laboratory (EECSL) at Korea University, School of Electrical Engineering [2]. The database contains time series vibration signals from electric motors under different conditions and faults. The data was collected by using microelectromechanical system capacitive accelerometers and Hall-effect sensors mounted onto the electric motor [1].

This study focuses on one out of the six different faults that was tested, misalignment, and compare it to the normal condition. This study uses the 60 Hz supply frequency operations.

The motors were tested under four different load conditions. A 30-kW DC generator was coupled to the test motors and was controlled at 0%, 50%, 75%, and 100% rated load by adjusting the field voltage [1].

4.1 Overview of healthy motor vibration signal

In this section, a healthy signal under steady-state conditions is analysed using the prototype sensor. The motor was powered via direct-on-line (DOL) supply. I downloaded the necessary files from the database [2] and cleaned up the data in MATLAB by removing NaN values, and normalized the time values by dividing them by the sampling frequency (800Hz). The data file contains X, Y, and Z-direction vibration signals. For the first part of this study, only X-axis data is considered. The signal amplitude was scaled to micrograms by using the scaling factor that can be found in the dataset. Figure 1 highlights an example of the raw vibration data.

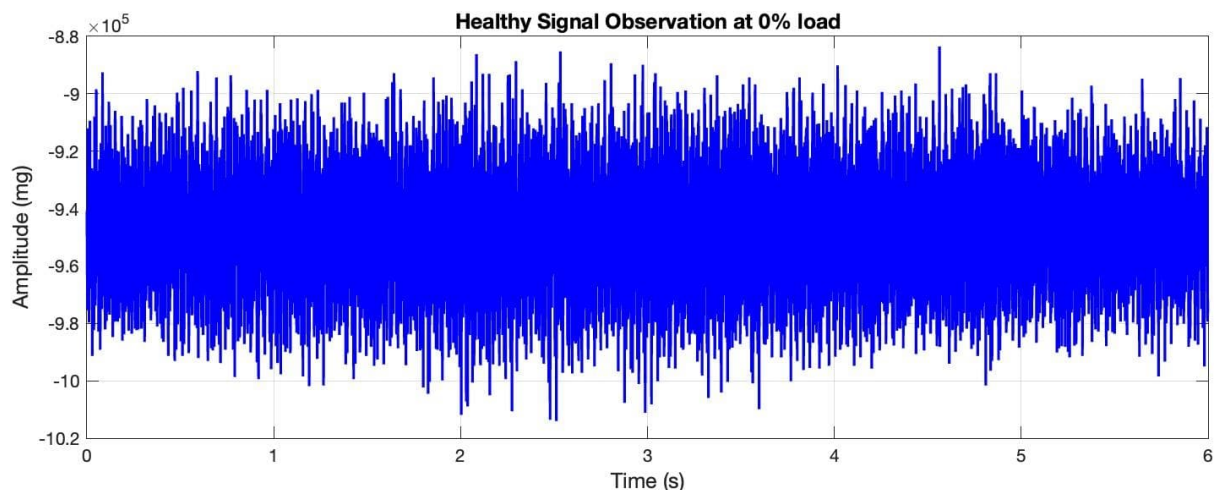


Figure 1: Vibration graphic of the healthy signal at 0% load.

For the Fast Fourier Transform (FFT), the frequency axis was created. According to the Nyquist-Shannon sampling theorem, to prevent misinterpretation of frequency components, the sampling frequency must be at least twice the bandwidth of the given signal. Conclusively, frequencies between 0 and 400 Hz are analysed. The frequency resolution is calculated as the sampling frequency divided by number of data points

(4800 data points per sample) which means it is approximately 0.1667 Hz. When applying the FFT and plotting the frequency axis produces the following figure.

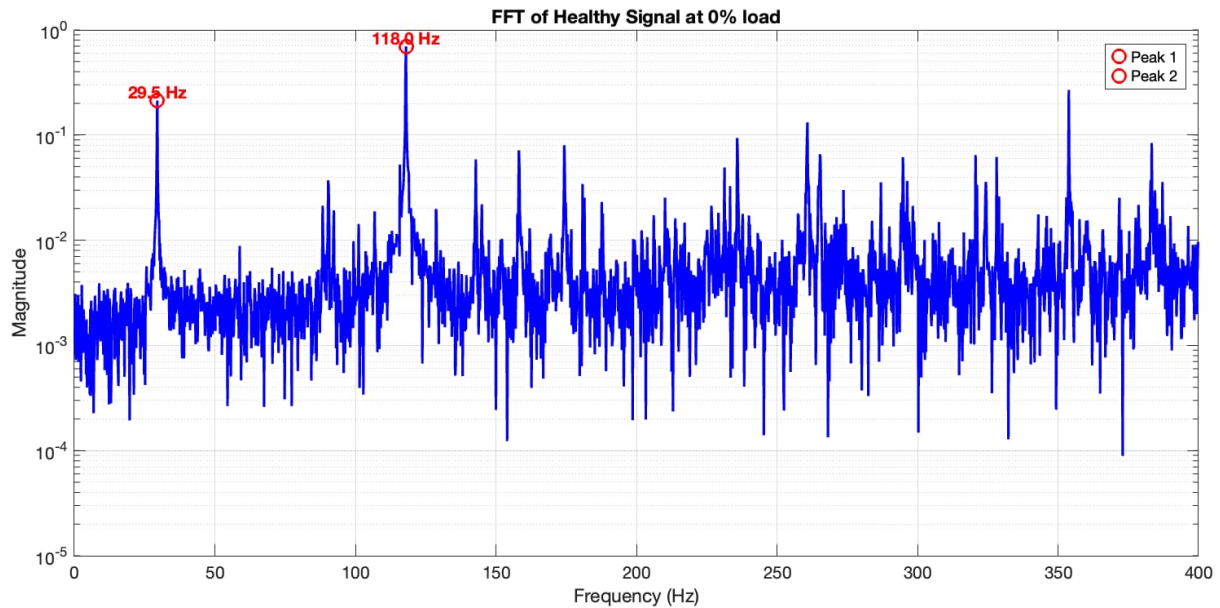


Figure 2: Frequency domain visualisation of the healthy signal at 0% load.

Figure 2 highlights the frequency spectrum of the signal of a healthy motor. The vertical axis of the graph is logarithmic, which means that in a linear graph, peaks would be even more noticeable, but this helps us gauge the overall signature of the signal. From the spectrum, the spectrum exhibits prominent peaks at around 30 Hz, 120 Hz, and just above 350 Hz.

When collecting the data, [1] used a motor with two pole pairs and a supply frequency of 60 Hz. This operating condition yields a speed of 1800 revolutions per minute for the motor, which in turn corresponds with the first peak around 30 Hz that can be observed in figure 2. Additional significant peaks appear at integer multiples of this mechanical frequency.

Figure three illustrates the differences between vibrations signals of the healthy motor under different loads.

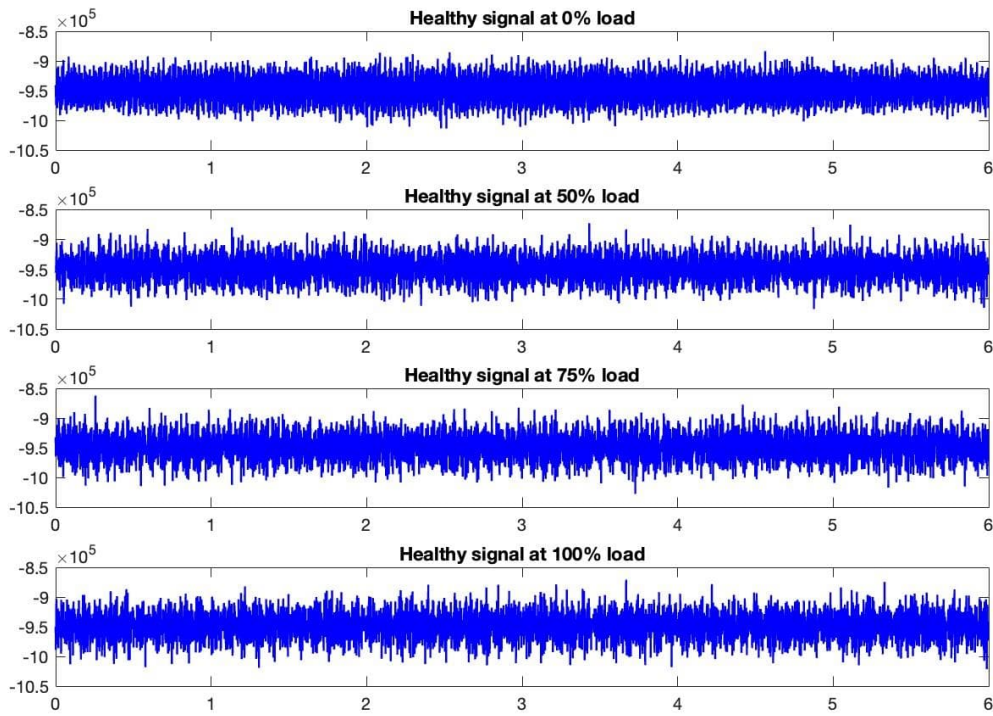


Figure 3: Time series view of healthy signal at varying load levels.

The signal at 0% load appears thicker than at other load levels. A lighter load could cause less mechanical dampening to the rotor's shaft, which could let it vibrate more freely. However, this can't be concluded with certainty.

When laying the four signals on top of each other as in figure 4, they exhibit similar characteristics in terms of frequency peaks. All signals share peaks around 30Hz and 120 Hz. The rest of the peaks seem more irregular and cannot be visually characterized as patterns.

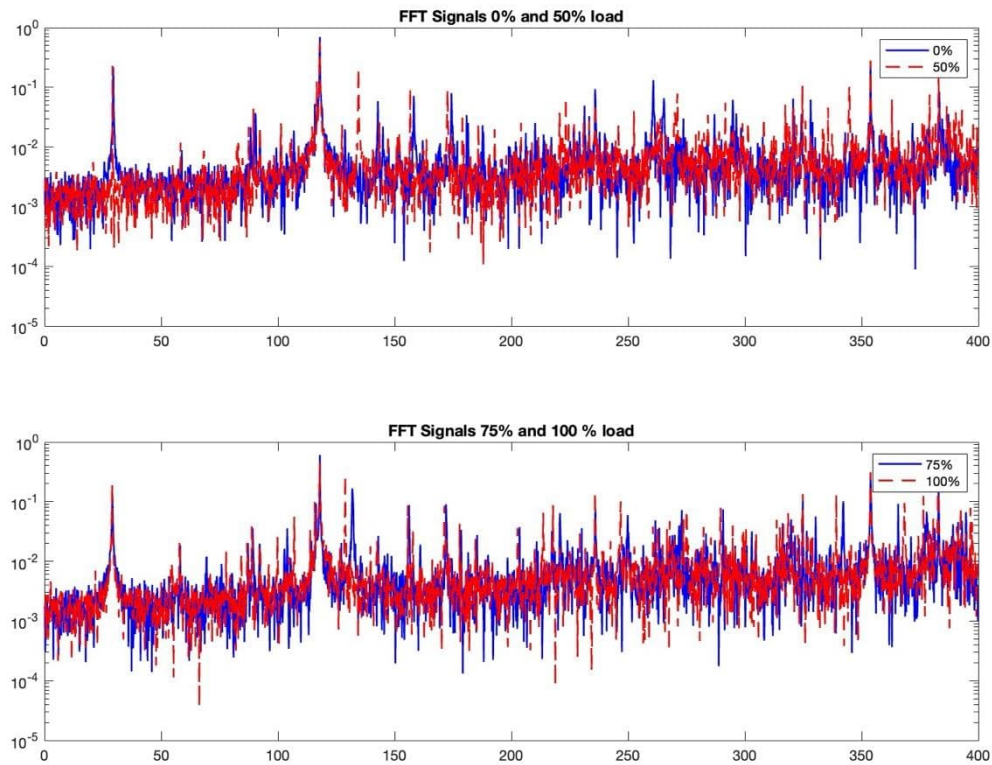


Figure 4: Comparison of FFT of healthy signals.

Let us verify the exact location and height of the two peaks that appear characteristic across all load levels, seen in figure 5.

Figure 5 confirms that the largest peak appears around 118 Hz for all signals. This is likely the fundamental frequency at which the electric motor vibrates, as it has significantly the highest magnitudes of 0.695, 0.562, 0.604, and 0.441, respectively, at 0%, 50%, 75% and 100% load, in comparison with the rest of the FFT.

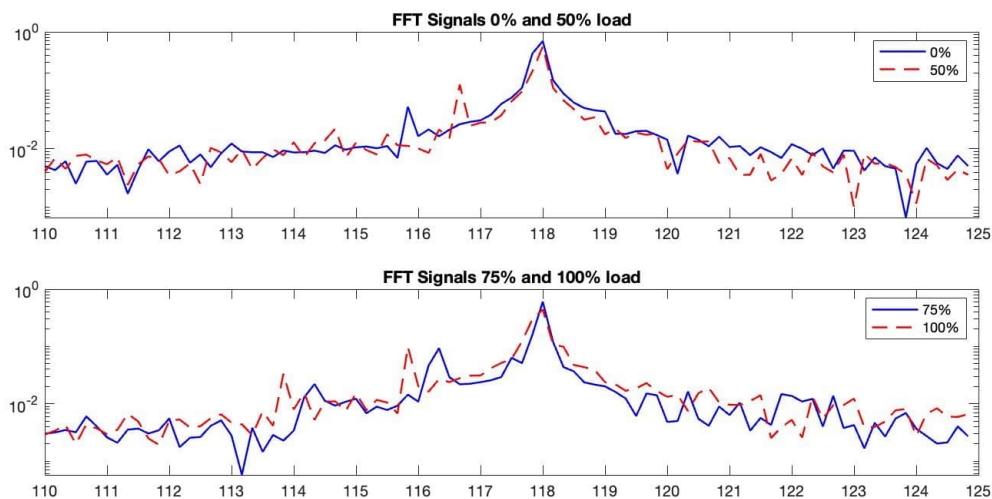


Figure 5: Zoomed-in view of the FFT of healthy signals around 120Hz.

The rotor speed frequency can be calculated with eq. (1). The constants mentioned in the formula can be found from [4], as it is also referenced in [1]. The experimental setup used two pole pairs and had a rotor slip of 0.75% and an AC of 60 Hz. Substituting these values into the formula yields

$$f_r = (1 - 0.75\%) \cdot \frac{60 \text{ Hz}}{2} \approx 29,775 \text{ Hz}$$

The exact location of these peaks falls in the range of 29.0 Hz to 29.5 Hz. This agrees with the position of the lower peak observed in figure 6 and most likely associated with the rotor speed frequency. These have magnitudes of 0.213, 0.228, 0.200, and 0.200, going from 0%, 50%, 75%, and 100% load.

According to [1], the Fourier transformation of the signal with a misaligned shaft is expected to exhibit an additional peak around 2 times rotor speed frequency, which is approximately 60 Hz.

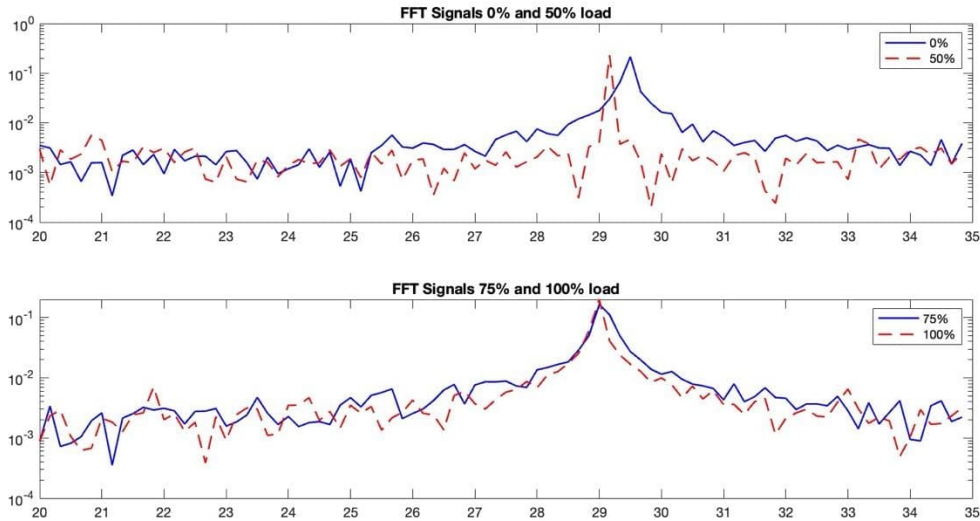


Figure 6: Zoomed-in view of the FFT of healthy signals around 30Hz.

4.2 Overview of misaligned motor vibration signal

For the faulty signals, I conducted the same signal processing and clean-up as for the healthy signals, which included removing NaN values and scaling the data by the scaling factor found in the data [2]. Figure 7 shows the raw vibration signal of faulty motor.

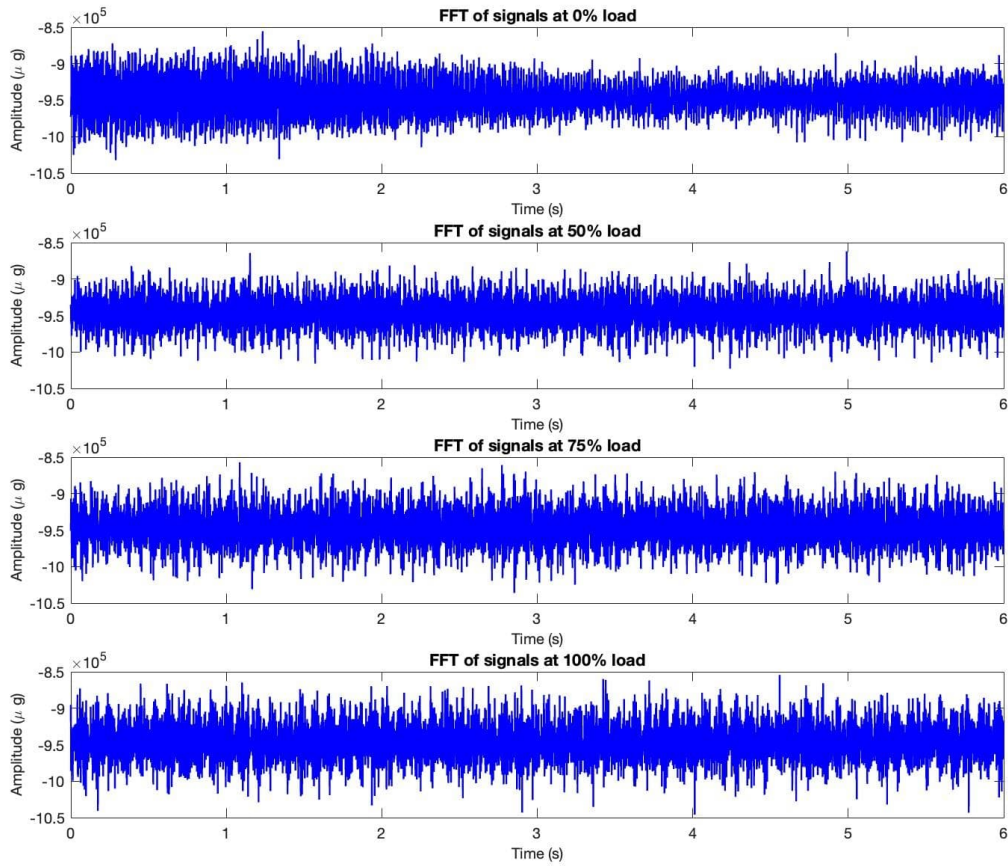


Figure 7: Time series view of 0.5 mm misalignment signal at varying load levels.

Simply viewing the raw signal, it is difficult to distinguish significant differences from the healthy signal. This is one example of how important tools like the Fourier transformation are for analyzing and finding differences in signals.

In figure 8, the faulty signal shows a greater amplitude in contrast to the healthy signal. This kind of behavior could explain why [11] found vibration analysis to be more effective for lighter loads.

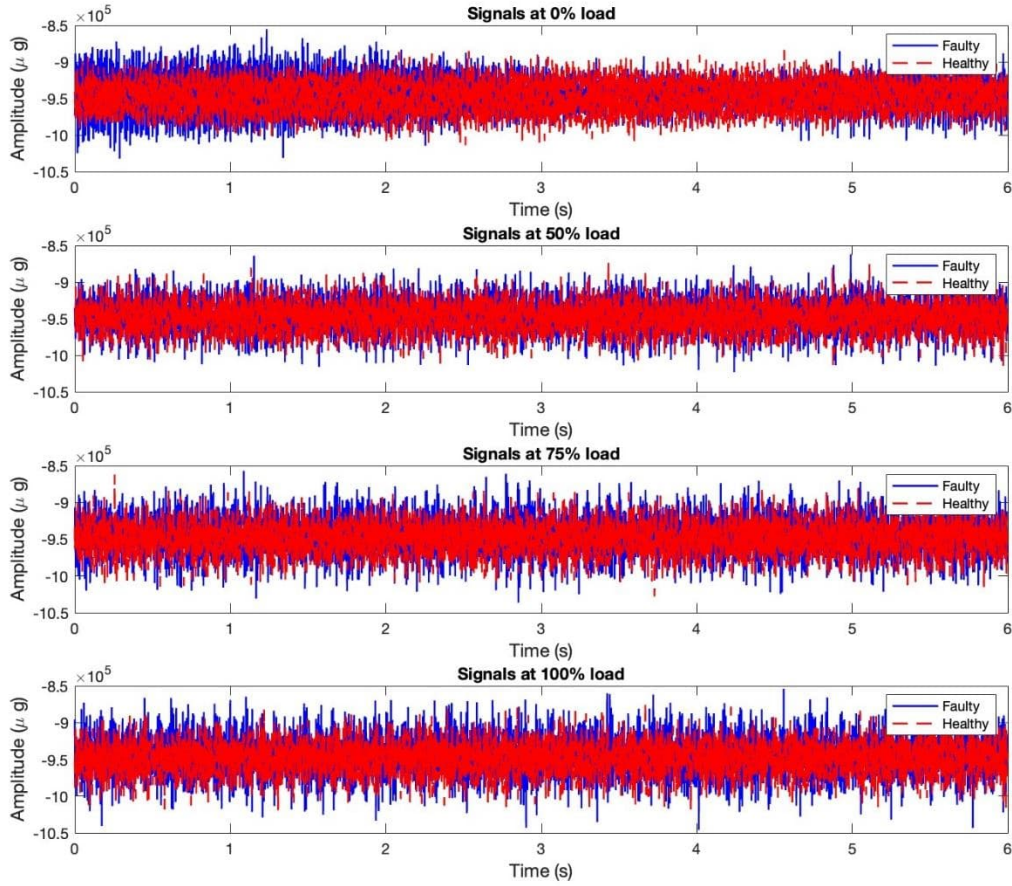


Figure 8: Time series view comparison of 0.5 mm misalignment signal and healthy signal at varying load levels.

In figure 9, the 1 mm misalignment results in greater deviation from the healthy signal, particularly at the 0% level of load. Overall, the 1 mm misalignment signal appears to have larger peaks that are further apart from each other than the thicker section of the vibration signal, which are not distinguishable to this degree in the 0.5 mm misalignment signal.

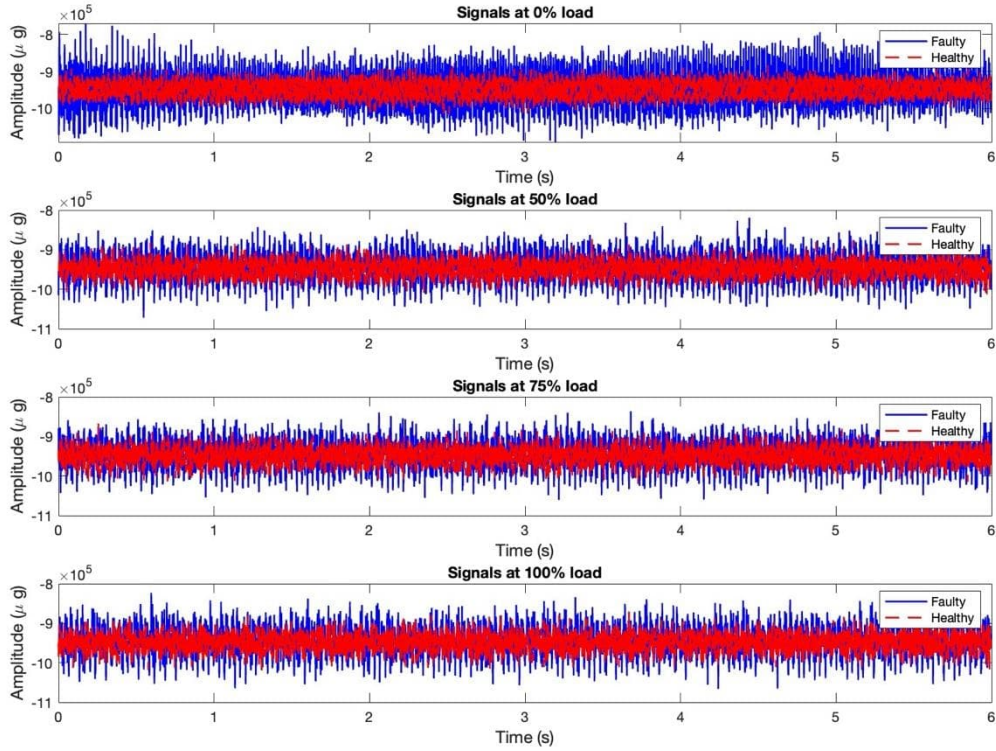


Figure 9: Time series view comparison of 1 mm misalignment signal and healthy signal at varying load levels.

Observing the frequency domain representation of the signal from the motor with 0.5 mm misalignment, figure 10 shows similar peaks at the beginning of the graph to the healthy motor. A larger peak around 120 Hz, and smaller peaks around 30 Hz. However, figure 10 shows a peak around 60 Hz, which is analysed further in this study.

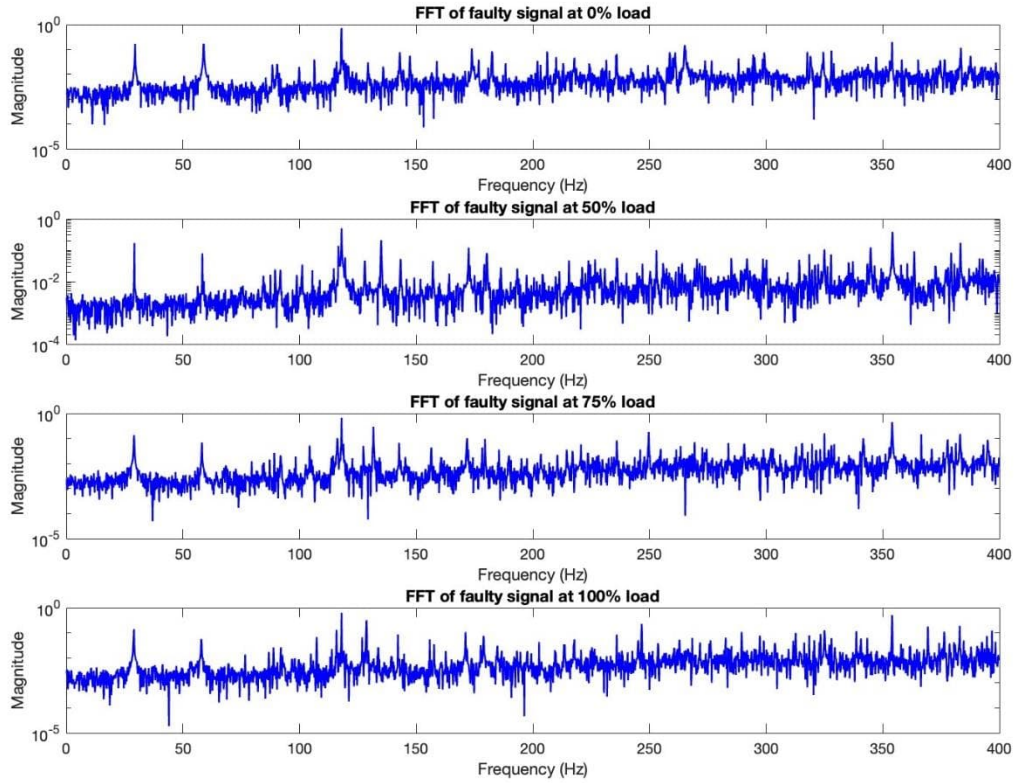


Figure 10: FFT frequency domain view of 0.5 mm misalignment signal at varying load levels.

Overlaying the FFT of 0.5 mm misalignment, figure 11 shows similarities in the frequency spectrum to the healthy vibration signal. On the spectrum of the faulty motor's signal, there appears a peak around 60 Hz. The exact location of the second peak in the faulty signal ranges from 57.83 Hz to 58.83 Hz. The theoretical location for the second peak would be 59.55 Hz. Multiplying the observed range for the rotor speed frequency by two gives us a range of 58 Hz to 59 Hz. These peaks have magnitudes between 0.17 and 0.6, decreasing as the load is increased. That gives us a confident reason to say that the peak near 59 Hz in the faulty motor signal would be the 2x harmonic of the rotor speed frequency, which matches the observation about peak multiples of the rotor speed frequency made in [1].

From these peaks, it is difficult to differentiate the 3 times rotor speed frequency and 4 times rotor speed frequency peaks mentioned in [11]. A clump of peaks can be observed slightly before 100 Hz, but the peak around 120 Hz, that was also present in the healthy signal, does not appear to be significantly larger in the faulty signal.

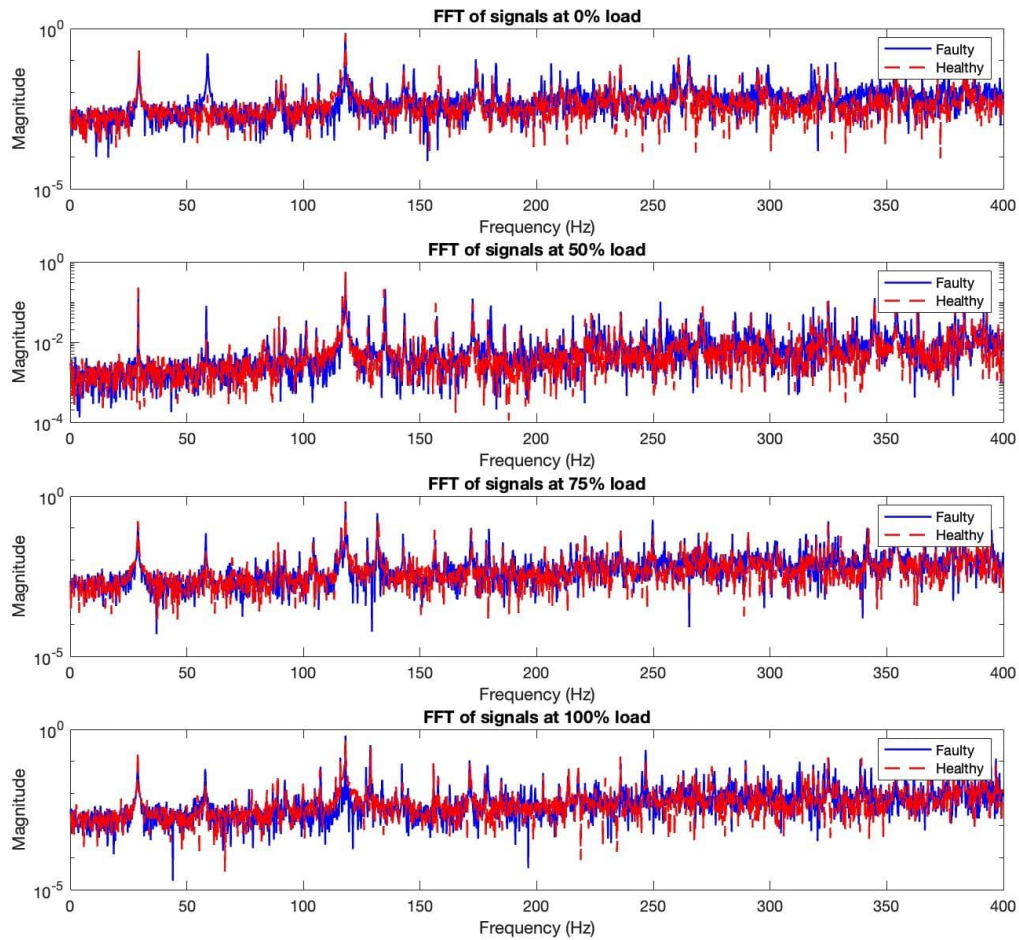


Figure 11: FFT frequency domain comparison of 0.5 mm misalignment and healthy signal at varying load levels.

In figure 12 for the 1 mm misalignment frequency domain, these same peaks around 60 Hz appear, with slightly greater magnitude falling in the range between 0.52 and 0.27, with the first peak being the largest. The exact location of the second peak also ranges between 58.0 Hz and 58.83 Hz, which is nearly identical to what is found on the 0.5 mm misalignment frequency spectrum. This would indicate that a greater level of misalignment tends to produce more deviating results from the characteristics of a healthy motor.

For the lightest load, figure 12 shows that the 1 mm misalignment creates significant peaks at around 150, 210, and 240 Hz. These are all near multiples of the rotor speed frequency that was calculated, but do not show up as clearly on the 0.5 mm misalignment. Comparing these harmonics of the rotor speed frequency to the higher loads, figure 12 shows that the peaks do not stand out much from the healthy signal. This follows the pattern of lighter loads creating more distinguishable characteristics from the healthy signal.

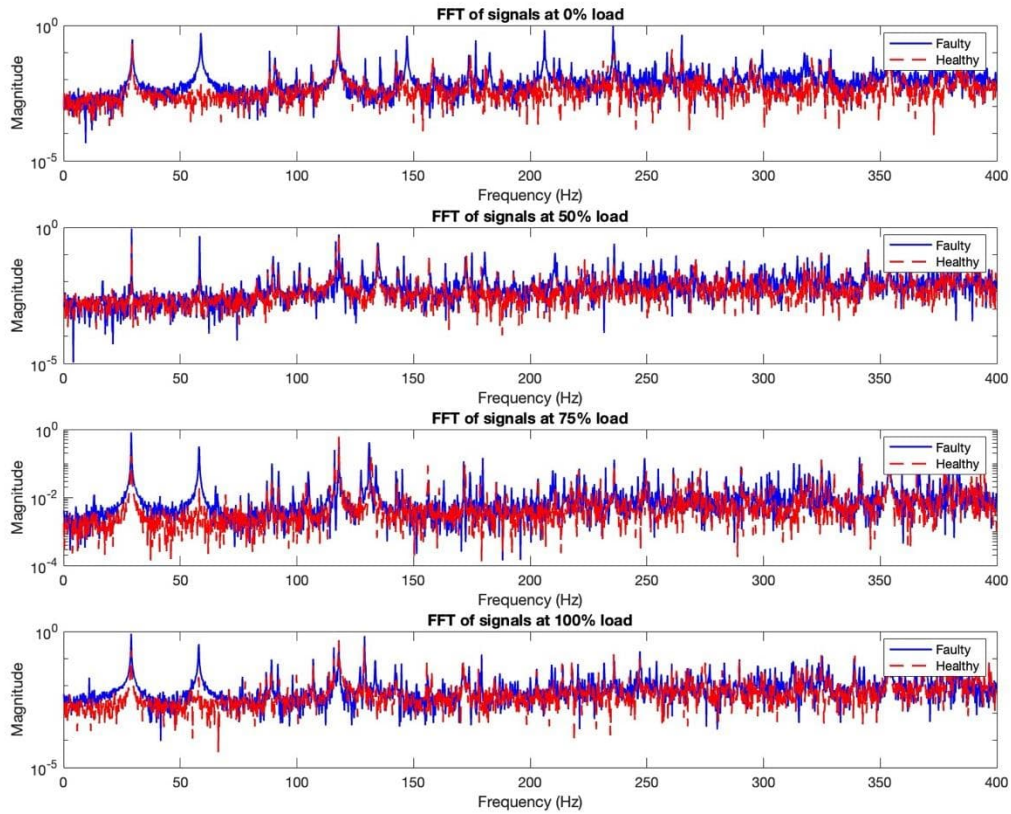


Figure 12: FFT frequency domain comparison of 1 mm misalignment and healthy signal at varying load levels.

4.3 Statistical comparison of time domain signals

To gauge the character of the time domain signals, statistical analysis was conducted on the characteristics of each signal. The built-in statistical functions of MATLAB were used to calculate the mean, median, skewness, kurtosis, and standard deviation of each signal and load level. Table 1 summarizes the basic statistical measures for the healthy and 0.5 mm misalignment vibration signal.

Table 1: Statistical comparison between healthy signal and 0.5 mm misalignment

Condition	Load level (%)	Mean	Median	Skewness	Kurtosis	Standard deviation (mg)
Healthy	0	0,0000	1.7190	-0.0864	2.1999	23.9943
0.5mm Misalignment	0	-0.000	-0.2406	0.0634	2.8376	25.2841
Healthy	50	-0.000	1.3406	-0.1359	2.8824	21.5351
0.5mm Misalignment	50	0.000	-0.6843	0.0093	2.9777	22.9230
Healthy	75	-0.000	0.8134	-0.1539	2.8957	22.9008

0.5mm Misalignment	75	-0.000	-1.6086	0.0799	2.9850	26.3983
Healthy	100	-0.000	1.2784	-0.2134	2.9755	22.9250
0.5mm Misalignment	100	-0.000	-1.2618	0.0835	3.0841	27.3203

These results show that the DC signal was effectively remove from the data, with all signals having insignificantly small means. These calculations show that the faulty signal has a slightly higher standard deviation, as well as slightly higher kurtosis but lower skewness.

Table 1 shows that the misalignment has a slightly higher standard deviation than the healthy signal. This corresponds with what can be observed in figure 8, where the 0.5 mm misalignment signal has a slightly higher amplitude in comparison to the healthy signal.

The kurtosis of both signals appears to be close to 3, which suggests the signal is relatively smooth without many outliers. However, it is notable that the absolute value of skewness is consistently lower for the faulty signals compared to the healthy ones. This may indicate changes in the signals' distribution that could be further explored.

Table 2: Statistical comparison between healthy signal and 1 mm misalignment.

Condition	Load level (%)	Mean	Median	Skewness	Kurtosis	Standard deviation (mg)
Healthy	0	0,0000	1.7190	-0.0864	2.1999	23.9943
1mm Misalignment	0	-0.000	-1.0093	0.1168	2.6478	50.8271
Healthy	50	-0.000	1.3406	-0.1359	2.8824	21.5351
1mm Misalignment	50	0.000	0.2481	-0.0735	2.7746	36.5079
Healthy	75	-0.000	0.8134	-0.1539	2.8957	22.9008
1mm Misalignment	75	-0.000	1.3138	-0.1236	2.7132	36.8684
Healthy	100	-0.000	1.2784	-0.2134	2.9755	22.9250
1mm Misalignment	100	-0.000	0.9670	-0.0897	2.7685	38.0035

Table 2 shows that the statistical analysis gives similar results for the 1 mm misalignment, except for the standard deviation being even larger than the 0.5 mm fault. This was expected, as figure 9 shows that the 1 mm misalignment caused a vibration signal with peaks further from the center than the healthy and 0.5 mm misalignment.

4.4 Calculation and analysis of spectral centroid

In this section, I compared the spectral centroid values for each signal. Spectral centroid is a way to measure the center of mass of a signal, typically used in audio signal processing. Recent studies, such as [9], have shown that observing changes in

the spectral centroid of a signal can be used to detect certain faults in mechanical machines.

The spectral centroid (SC) values were calculated for each load level with MATLAB using eq. (2). The MATLAB code for this can be found in the appendix. It yielded the following results for the X-axis data:

Table 3: Comparison of spectral centroid values between healthy, 0.5 mm misalignment, and 1 mm misalignment in the X-direction.

Load level (%)	SC Healthy (Hz)	SC Faulty 0.5 mm (Hz)	SC Faulty 1 mm (Hz)
0	45.67	58.56	80.68
50	52.59	62.84	69.45
75	57.10	65.05	69.43
100	59.22	65.33	72.73

Table 3 summarizes the SC values for the healthy, 0.5 mm misalignment, and 1 mm misalignment signals.

Notably, the SC values for the larger misalignment are further away from the healthy signal than the smaller misalignment. This corresponds with the intuitive expectation of a bigger misalignment causing more chaotic movement or vibration for the motor.

In contrast to the healthy and 0.5 mm misalignment motor, the SC for the 1 mm misalignment motor is higher at 0% than the other measurements. This could be due to transient vibrations from starting the motor under less load. However, this remains speculative without further measurements and is something that could be further explored.

Figures 13 and 14 summarize the position of the SC in relation to the whole FFT spectrum of healthy and faulty signals. The graphs confirm the larger difference of SC value between the healthy and 0.5 mm misalignment compared to the healthy and 1 mm misalignment.

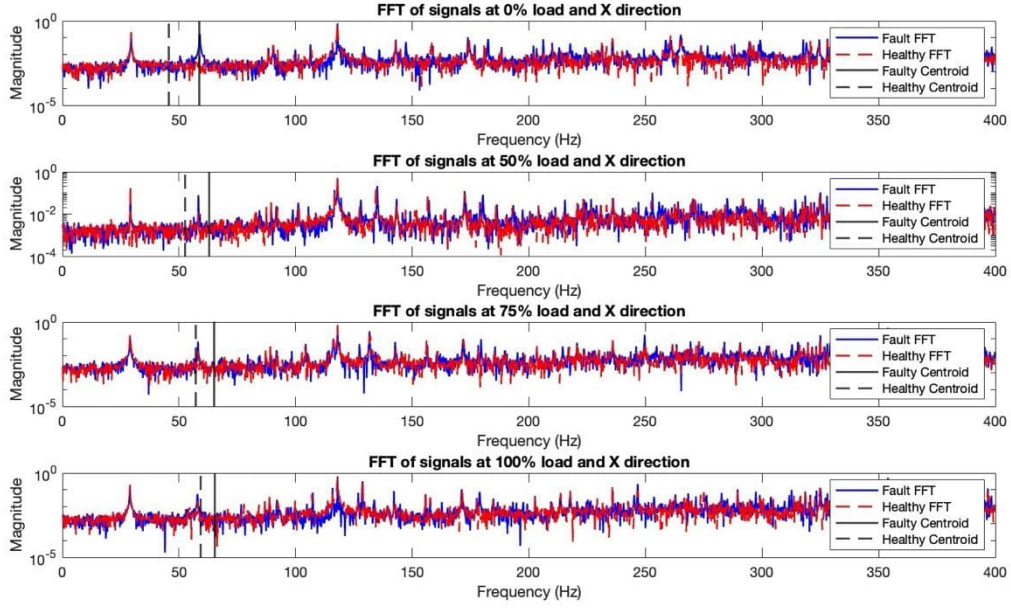


Figure 13: Spectral centroid value in comparison to the FFT of healthy and 0.5 mm misalignment signal.

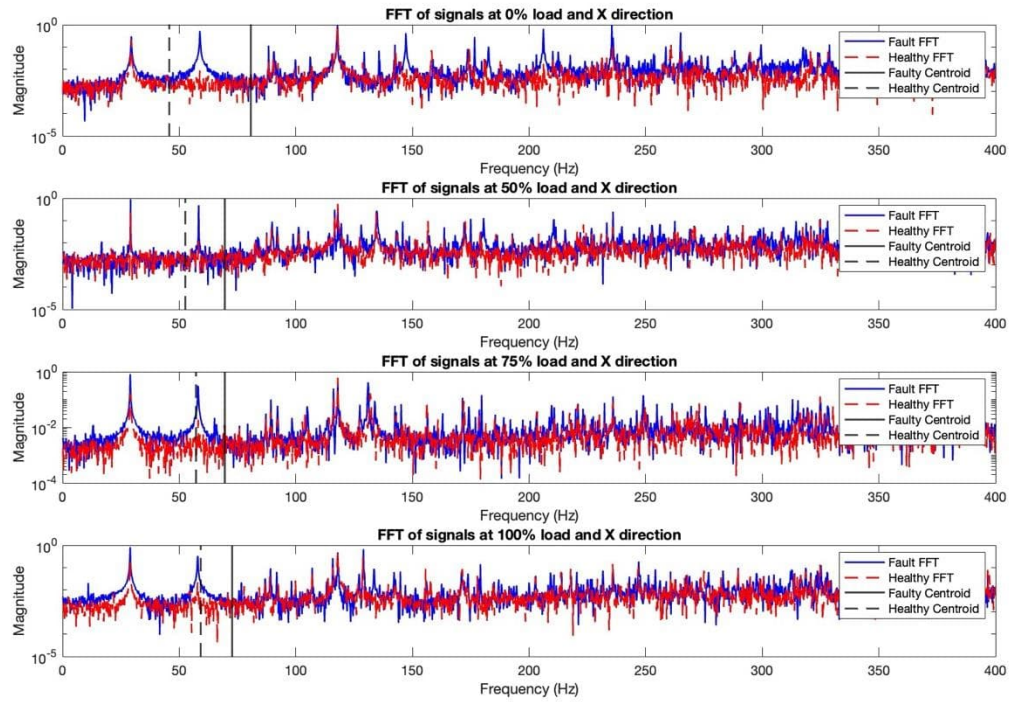


Figure 14: Spectral centroid value in comparison to the FFT of healthy and 1 mm misalignment signal.

Comparing the X-axis data to the other two axes gives some interesting results. The SC values for Y- and Z-axis data are larger by almost a factor of two, but the differences between SC values are significantly smaller.

Table 4: Comparison of spectral centroid values between healthy and 0.5 mm misalignment in the Y- and Z-directions.

Load level (%)	SC Healthy Y-axis (Hz)	SC Faulty Y-axis 0.5 mm (Hz)	SC Healthy Z-axis (Hz)	SC Faulty Z-axis 0.5 mm (Hz)
0	145.4180	170.6702	103.4256	116.7251
50	155.6571	162.3545	130.7368	138.4146
75	152.8425	160.0567	139.6815	139.1385
100	154.9958	161.6470	140.7031	136.6680

Figure 15 shows the location of the SC for the healthy and 0.5 mm misaligned signal in relation to the rest of the frequency axis.

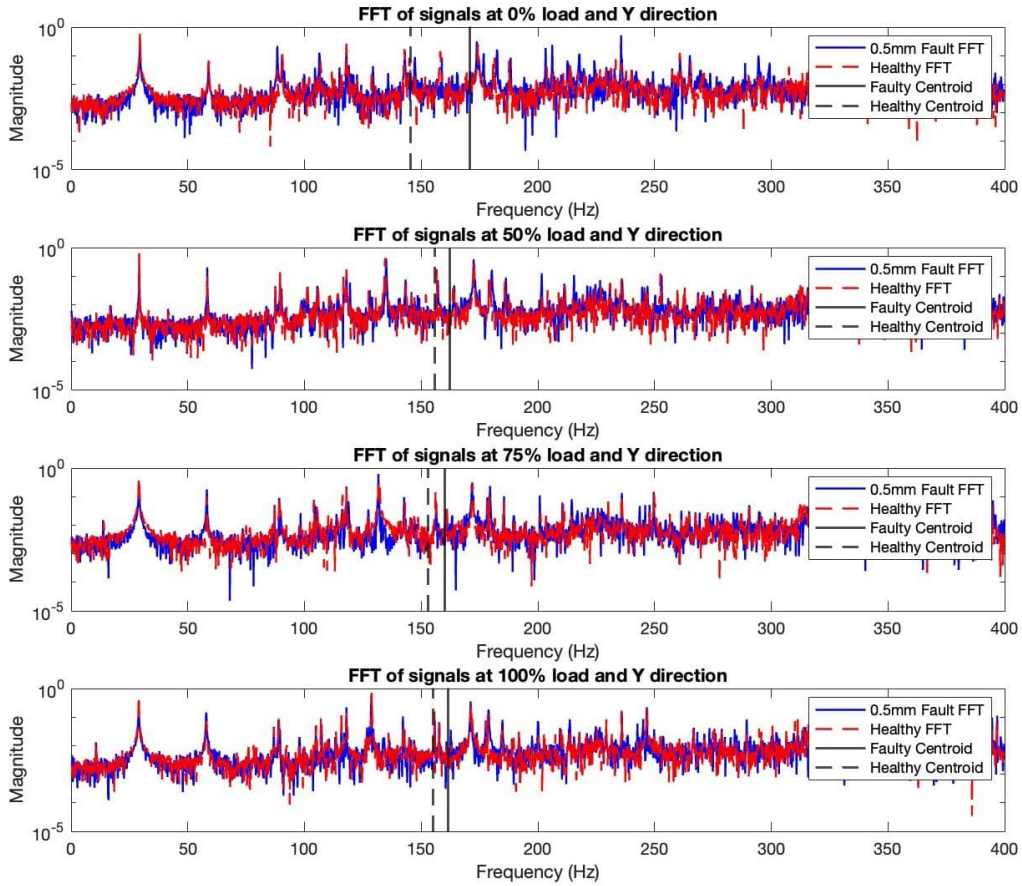


Figure 15: Spectral centroid value in comparison to the FFT of healthy and 0.5 mm misalignment signal in the Y-direction.

In contrast to the X-axis, the Y-axis for the healthy IM appears to have a similar peak around the $2f_r$ position, similar to the misaligned motor, with a similar magnitude, shown in figure 15. This kind of behavior is associated with misalignment, as mentioned in [1]. Such peaks could be caused by background vibrations or sensor positioning, but further investigation is needed to determine the cause. Despite this, the trend of SC being higher for the misalignment persists with the loads tested, albeit

the differences in SC are smaller between healthy and misaligned IMs in comparison to the X-axis.

Figure 16 shows that the Z-axis does not contain the unusual characteristic of the healthy motor having a peak around 2 times rotor speed frequency. However, the trend of SC being larger ends, as SC for the healthy motor is larger with 100% load in comparison to the misaligned motor.

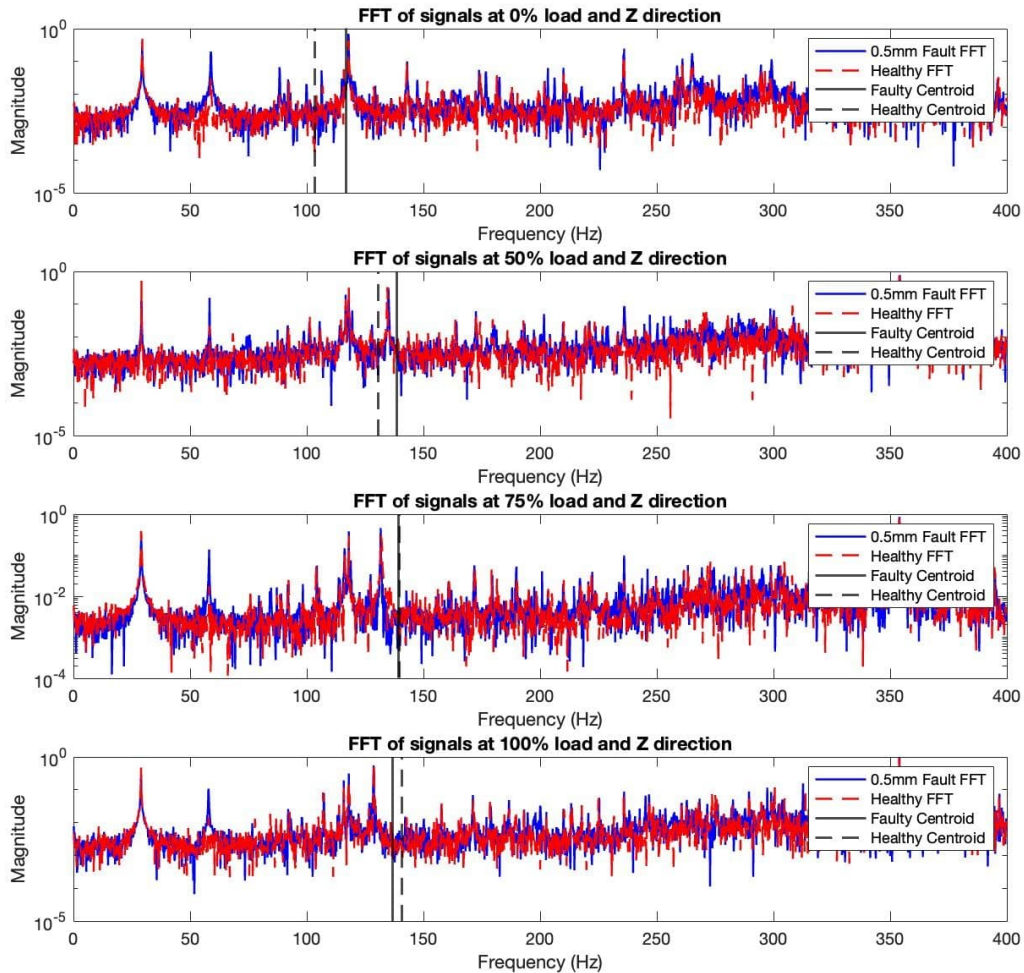


Figure 16: Spectral centroid value in comparison to the FFT of healthy and 0.5 mm misalignment signal in the Z-direction.

The SC values for the 1 mm misalignment in Y- and Z-axis directions were calculated using the same MATLAB code that was used for 0.5 mm misalignment SC values. The code can be found in the appendix.

Table 5: Comparison of spectral centroid values between healthy and 1 mm misalignment in the Y- and Z-directions.

Load level (%)	SC Healthy Y-axis (Hz)	SC Faulty Y-axis 1 mm (Hz)	SC Healthy Z-axis (Hz)	SC Faulty Z-axis 1 mm (Hz)
0	145.4180	175.1904	103.4256	135.1908

50	155.6571	168.6910	130.7368	137.2314
75	152.8425	155.1730	139.6815	138.4335
100	154.9958	157.7356	140.7031	140.2466

Table 5 summarizes the SC values for the 1 mm misalignment fault in the Y and Z axes. The calculations show a trend of the misalignment having higher SC values in comparison to the healthy signal continues, but the 1 mm misalignment values are not as distinctively different from the 0.5 mm misalignment SC values as in the X-axis comparison.

Figure 17 and figure 18 show that the black dashed line representing the SC value of a healthy motor is closer to the solid black line representing the SC values of the faulty motor. At certain load levels, the dashed line surpassing the solid line. This shows that a single axis of data may not be viable to detect differences between misaligned and healthy motor.

The 2 times rotor speed frequency factor observed in the X-direction also appears in the Y- and Z-direction for the 1 mm misalignment as a significant peak, characteristic of motors with misalignment type fault [1].

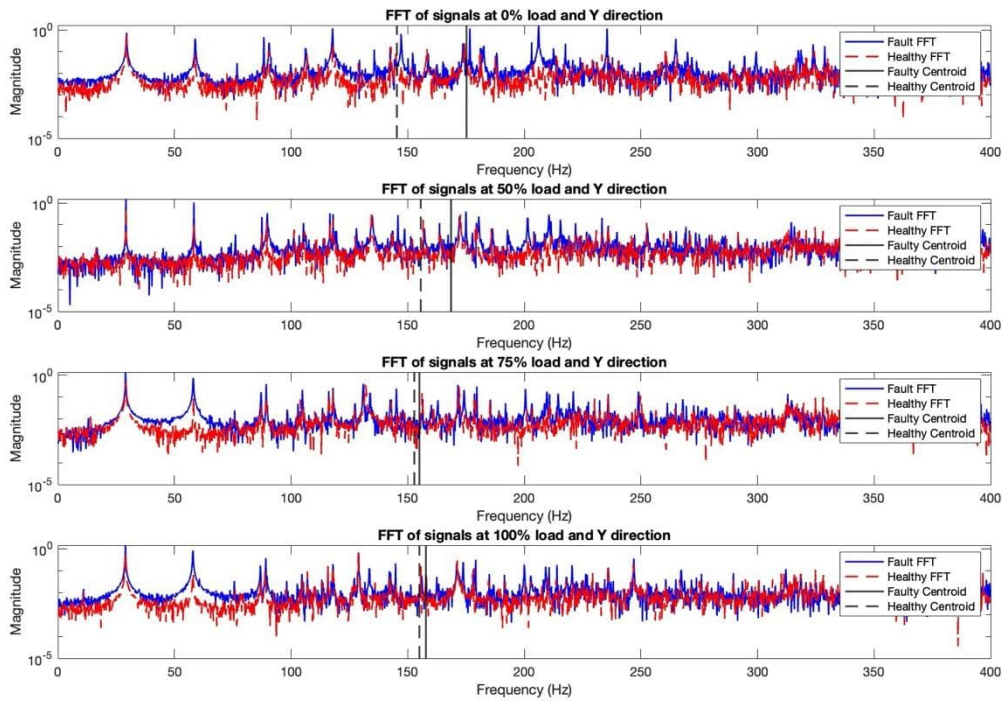


Figure 17: Spectral centroid value in comparison to the FFT of healthy and 1 mm misalignment signal in the Y-direction.

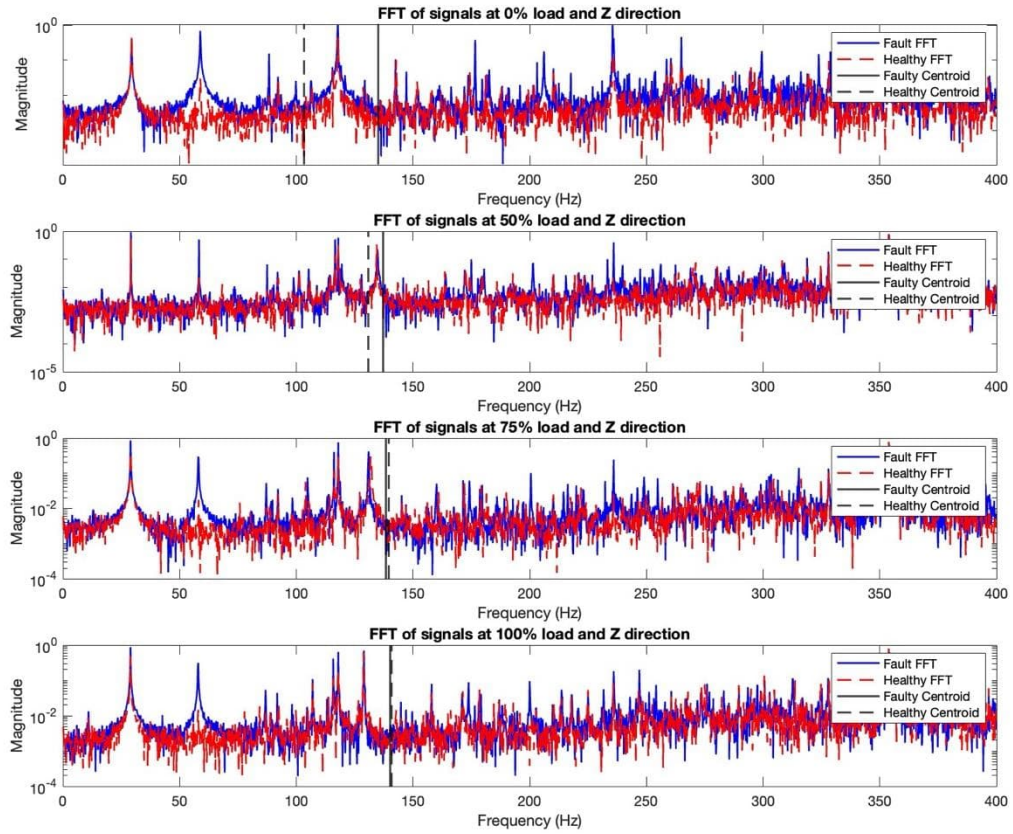


Figure 18: Spectral centroid value in comparison to the FFT of healthy and 1 mm misalignment signal in the Z-direction.

5 Research results

5.1 Spectral centroid tests

This study explored the use of spectral centroid (SC) as a fault indicator for induction motors (IMs) with shaft misalignment type of faults. SC values were calculated for all three axes of movement (X, Y, Z), under healthy, 0.5 mm misalignment, and 1 mm misalignment conditions, under four different resistance levels (0%, 50%, 75% and 100%).

5.1.1 X-axis result

The SC values showed a consistent trend of increasing with the severity of misalignment. At 0% load, the SC rose from 45.67 Hz for the healthy motor to 58.56 Hz for the 0.5 mm misalignment to 80.68 Hz for the 1 mm misalignment. This trend was observed across all load levels. Both the healthy motor and the 0.5 mm misalignment showed a trend of increasing SC values as the resistance increased. The 1 mm misalignment started at 80.68 Hz with 0% resistance, dropped down to between 69.43-69.45 Hz with 50V and 75V resistance, but rose to 72.73 Hz with 100% load. Increasing the load decreased the difference in SC between healthy and misaligned. At its smallest, with 100% load, the difference between the healthy and misaligned motor was 6.11 Hz.

5.1.2 Y- and Z-axis results

The tests in the Y- and Z-axis directions were more inconsistent than what was observed in the X-axis. Additionally, there was a clear shift in the magnitude of their SC from the X-axis. The Y-direction had SC values between 145 and 175 Hz, while the Z-direction had SC values between 103 and 141 Hz.

In the Y-direction, the trend of misalignments having larger SC than the healthy motor persisted, but the 1 mm misalignment did not consistently produce a SC bigger than the 0.5 mm misalignment. The difference in SC between healthy and misaligned motors became smaller again, the bigger the load used. At its smallest, with 100% load, the difference was 2.74 Hz. With 0% load, the difference in SC was 25.25 Hz for the 0.5 mm misalignment to the healthy motor and 29.77 Hz for the 1 mm misalignment to the healthy motor.

In the Z-direction, the trend of misaligned motors having a larger SC than the healthy motor diminished, as the healthy motor had a larger SC than both misaligned motors. It was not by a comparatively large amount, being larger by 0.45-4.04 Hz, with most differences closer to the 0.45 Hz difference than the 4.04 Hz difference.

Even if the differences in SC were less pronounced or consistent in the Y- and Z-axes than the X-axis, these findings support the idea of misalignment faults causing a redistribution in vibrations towards higher frequencies.

5.2 Statistical analysis of the time domain vibration signal

To gauge the characteristics of the vibration signal, basic statistical analysis was done for each time domain signal. The DC component was removed from all signals to eliminate bias. These tests were only done on the X-axis data.

The tests showed a consistent pattern of misaligned motors having a larger standard deviation than the healthy motor. The healthy motor had a standard deviation ranging between 21.5 and 24.0 mg, while the 0.5 mm misalignment caused a standard deviation ranging between 22.9 and 27.4 mg. The 1 mm misalignment had a standard deviation ranging between 36.8 and 50.9 mg, with the largest standard deviation occurring at the 0% load.

The values for skewness and kurtosis remained relatively stable across the board. Skewness ranging between -0.22 and 0.11, and kurtosis between 2.1 and 3.0.

5.3 Frequency domain verification

The study done by [1] mentioned frequencies that come up as larger peaks in the FFT of a faulty motor in comparison to a healthy motor. For the misalignment fault, the sensitive indicator frequency was mentioned to be the rotor speed frequency itself, and particularly a larger factor at twice the rotor speed frequency, with smaller peaks at integer multiples. The rotor speed frequency was calculated theoretically to be 29.78 Hz. On the FFT, a clear peak appeared in the range between 29.0 and 29.5 Hz, varying between loads. Analyzing the FFT of the misaligned motor, a clear peak appeared between 57.83 Hz and 58.83 Hz, having a magnitude between 0.17 and 0.6 for the 0.5 mm misalignment and 0.27 and 0.52 for the 1 mm misalignment. This confirms the pattern observed in the earlier study and correlates with the shift in spectral centroid.

On the 1 mm misalignment, some of the other integer multiples of the rotor speed frequency showed up as distinguishable peaks but were only significant on the lightest load. For higher loads, the integer multiples did not stand out from the healthy signals' frequency domain values.

Additionally, the Y-axis for the healthy IM showed an unusual characteristic of containing a similar peak around 2 times rotor speed frequency to the motors with misalignment faults. This characteristic showed up only in the Y-axis of the healthy motor and is something that could be further researched.

5.4 Discussion

This section discusses the effectiveness of SC as a fault indicator for detecting misalignment faults in IMs. Overall, the results signified a trend of SC being higher at lighter loads for motors with misalignment. However, there were certain characteristic differences between different axes of measurement.

One potential reason for the X- and Y-axes having more significant differences between healthy and misaligned motors than the Z-axis could be if the Z-axis was in the same direction as the shaft itself. Intuitively, a parallel misalignment would more likely cause vibrations across the shaft rather than along the shaft, but this remains speculative because the orientation of each axis label was not specified [1].

SC has some inherent flaws as a fault detection method for misalignment. SC calculates the average frequency of a signal. It does not characterize the behavior or outlook of a signal. Given the results from this study, which showed that misalignment showed a trend of causing a peak around 2 times rotor speed frequency, other statistical methods could prove more effective at detecting misalignment faults in comparison to SC.

In addition, the frequency domain signal appeared to have some noise towards the higher frequencies observed in this study. Using a filtering method for either the frequency spectrum or the time domain signal could highlight clearer differences between signals.

5.5 Reliability analysis

The findings made in this study are limited by a few factors, one of them being the controlled environment in which the data were collected. Real-world environments may introduce noise and unexpected variables to the data collected, which can severely affect the reliability of the patterns found in controlled environments. It cannot be stated that the patterns found would sustain under more varying load conditions.

Moreover, the data used a limited sample size. The study contained only a single motor for each condition: healthy, 0.5 mm misalignment, and 1 mm misalignment. Observing patterns over larger data sets with multiple different motors would give more reliable and rigorous results.

6 Conclusions

This study aimed to explore the use of spectral centroid (SC) as a fault detection method for parallel misalignment in the induction motor (IM) shaft. Vibrations of the motors were measured in all three axes of movement, and signals were processed from the time domain to the frequency domain. SC calculates the centre of mass for the frequency spectrum and could show a redistribution of frequencies in the vibration signal of misaligned IMs.

The results showed a clear pattern in the X-axis of SC having consistently higher values in the misaligned IM compared to the signal of a healthy IM. However, the differences became smaller as the load was increased, suggesting that SC may be more effective under lighter loads.

The Y- and Z-axis had less consistent results. The Y-axis had the same trend as the X-axis of SC being higher for the misaligned IMs, but the differences became even less significant. For the Z-axis, the results became unpredictable. For higher loads, the SC for healthy IM was larger than that of the misaligned IM. For lighter loads, both in the Y- and Z-axis, the SC stayed consistently higher for misaligned IM in comparison with healthy IM. These findings highlight that SC can be sensitive both in terms of load and direction and may not be able to consistently detect misalignment faults under varying conditions or loads.

These findings suggest that SC has potential for detecting misalignment faults under lighter conditions, but the small data set may limit generalisation. Only one healthy IM was tested under each of four load conditions, and only two different degrees of parallel misalignment (0.5 mm and 1 mm) were tested under each of the four load conditions.

In conclusion, spectral centroid may serve as a fault detection method for IMs under lighter loads. Since the spectral centroid can produce varying results between different axes of movement, all axes should be captured and analysed for reliability. These findings were done under limited conditions, and further research is necessary to ensure the robustness of the method. Further work could include analysing data from multiple motors under varying conditions, including real-world settings.

7 Recommendations

This study focused on the vibration-based fault detection of IMs using spectral centroid values. While the methods used showed some promise, several key areas could be expanded and improved upon, particularly to increase robustness and breadth of data.

7.1 Expand to real-world environments

The data used in this study were collected under controlled laboratory conditions to control the variables and types of faults. While this is useful for isolating faults from each other, real-world conditions often introduce noise and other external factors that can significantly affect the vibration signal. Therefore, it would be beneficial to conduct the same kind of experiments on data collected in dynamic real-world environments, such as factories or industrial complexes.

7.2 Broaden motor data and fault types

One significant limiting factor of this study is the amount of data used. This study uses the data from one motor under a limited number of conditions. While there was data for VFD conditions, only DOL data was used to stay within the scope. The first step to expanding this study would be to include the VFD data in the experiments. In addition, analyzing data from a larger number of motors from different manufacturers and different conditions would radically increase the robustness of findings.

The use of SC for detecting faults could be expanded to different kinds of faults. This study focused on misalignment fault type, but exploring the use of SC values for other kinds of fault types is worth considering.

7.3 Investigating more advanced analytical approaches

This study focused on relatively simple statistical measures for detecting faults. More complex methods could be applied not only in signal processing but also for analyzing the signal. Methods like filtering or windowing could give more consistent and better-quality data, from which it could be easier to recognize patterns. Tools like machine learning and artificial intelligence could prove useful for finding patterns that otherwise are difficult to recognize with the mathematics used in this study.

In addition, combining such more advanced detection methods of the vibration signal with other kinds of approaches, such as MCSA, could give a more complete picture of the behavior and characteristics of a faulty motor.

More advanced statistical methods could reveal clearer patterns in the behavior between healthy and misaligned motors.

7.4 Developing open datasets and benchmarks

One barrier to progress in this area is the lack of openly available data about IMs. Developing a platform where vibration and current data could be published from

different environments, faults, and motor types could rapidly accelerate the research for patterns of faulty motors. Broader access to diverse datasets could enable researchers to compare different statistical and signal processing methods and collaboratively expand the understanding of reliable indicators.

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Appendix

Code 4.1 Overview of healthy signal data

```
volt = [0, 50, 75, 100]; % Different load levels to the motor

signals = cell(1, length(volt)); % Preallocate cell arrays
times = cell(1, length(volt));
ffts = cell(1, length(volt));

k = 0; % index

% Processing all four files
for i = 1:4
    filePath = sprintf('Users/joakimeklund/Downloads/1 Prototype
healthy/DOL/Healthy_DOL60hz_%d_vib.txt', volt(i));

    data = readmatrix(filePath);
    data = data(~any(isnan(data), 2), :); % Remove NaN rows

    k = k + 1;

    % Extract time and signal columns, and correcting them
    times{k} = data(:,1)/800;
    signals{k} = data(:,2)*0.061*1000;

    N = length(signals{k});
    Y = fft(signals{k})/N;
    ffts{k} = abs(Y(1:N/2))*0.061000/1000; % Taking only positive frequencies
end

Fs = 800; % Sampling frequency
f = Fs * (0:(N/2)-1) / N; % Frequency axis
N1 = length(f) % Checking length of frequency axis

% Raw healthy signal at 0%
figure;
plot(times{1}, signals{1}, 'b', 'LineWidth', 1.5, 'LineStyle','-');
xlabel('Time (s)', 'FontSize', 12);
ylabel('Amplitude (\mu g)', 'FontSize', 12);
title('Healthy Signal Observation at 0% load', 'FontSize', 14);
grid on;

% FFT of healthy signal at 0%
figure;
semilogy(f(2:end), ffts{1}(2:end), 'b', 'LineWidth', 1.5); % Removing the DC component by skipping the
first FFT value
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title(sprintf('FFT of Healthy Signal at %d%% load', 0));
```

```

grid on;
legend(sprintf('FFT Spectrum %d%% load', 0));

% All raw signals at once
figure;
for i = 1:4
    subplot(4, 1, i);
    plot(times{i}, signals{i}, 'b', 'LineWidth', 1.2);
    hold on;
    xlabel('Time (s)', 'FontSize', 10);
    ylabel('Amplitude (\mu g)', 'FontSize', 10);
    title(sprintf('Raw signal at %d%% load', volt(i)));
end

% Overlaying FFTs on top of each other, healthy signal
figure;
subplot(2, 1, 1);
semilogy(f(2:end), ffts{1}(2:end), 'b', 'LineWidth', 1.2);
hold on;
semilogy(f(2:end), ffts{2}(2:end), 'r-', 'LineWidth', 1.2);
hold off;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('0%', '50%');
title('FFT Signals 0% and 50% load');

subplot(2, 1, 2);
semilogy(f(2:end), ffts{3}(2:end), 'b', 'LineWidth', 1.2);
hold on;
semilogy(f(2:end), ffts{4}(2:end), 'r-', 'LineWidth', 1.2);
hold off;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('75%', '100%');
title('FFT Signals 75% and 100% load');

% Zoomed in view around 120 Hz
figure
subplot(2, 1, 1);
semilogy(f(661:750), ffts{1}(661:750), 'b', 'LineWidth', 1.2);
hold on;
semilogy(f(661:750), ffts{2}(661:750), 'r-', 'LineWidth', 1.2);
hold off;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('0%', '50%');
title('FFT Signals 0% load and 50% load');
xticks(110:1:125);

subplot(2, 1, 2);

```

```

semilogy(f(661:750), ffts{3}(661:750), 'b', 'LineWidth', 1.2);
hold on;
semilogy(f(661:750), ffts{4}(661:750), 'r-', 'LineWidth', 1.2);
hold off;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('75%', '100%');
title('FFT Signals 75% and 100% load');
xticks(110:1:125);

% Zoomed in view around 30 Hz
figure;
subplot(2, 1, 1);
semilogy(f(121:210), ffts{1}(121:210), 'b', 'LineWidth', 1.2);
hold on;
semilogy(f(121:210), ffts{2}(121:210), 'r-', 'LineWidth', 1.2);
hold off;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('0%', '50%');
title('FFT Signals 0% and 50% load');
xticks(20:1:35);

subplot(2, 1, 2);
semilogy(f(121:210), ffts{3}(121:210), 'b', 'LineWidth', 1.2);
hold on;
semilogy(f(121:210), ffts{4}(121:210), 'r-', 'LineWidth', 1.2);
hold off;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
legend('75%', '100%');
title('FFT Signals 75% and 100% load');
xticks(20:1:35);

```

Code 4.2 Overview of faulty signal

```

%% Plotting

% All raw misalignment signals at once
figure;
for i = 1:4
    subplot(4, 1, i);
    plot(times_f{i}, signals_f{i}, 'b', 'LineWidth', 1.2);
    hold on;
    xlabel('Time (s)', 'FontSize', 10);
    ylabel('Amplitude (\mu g)', 'FontSize', 10);
    title(sprintf('FFT of signals at %d%% load', volt(i)));
end

% Healthy on top of faulty raw

```

```

figure;
for i = 1:4
    subplot(4, 1, i);
    plot(times_f{i}, signals_f{i}(1:end), 'b', 'LineWidth', 1.2);
    hold on;
    plot(times_h{i}, signals_h{i}(1:end), 'r--', 'LineWidth', 1.2);
    hold off;

    xlabel('Time (s)', 'FontSize', 12);
    ylabel('Amplitude (\mu g)', 'FontSize', 12);
    legend('Faulty', 'Healthy');
    title(sprintf('Signals at %d%% load', volt(i)));
end

%FFT of faulty signal
figure;
for i = 1:4
    subplot(4, 1, i);
    semilogy(f(2:end), ffts_f{i}(2:end), 'b', 'LineWidth', 1.2);
    xlabel('Frequency (Hz)');
    ylabel('Magnitude');
    title(sprintf('FFT of faulty signal at %d%% load', volt(i)));
end

% Healthy signal FFT on top of faulty signal FFT
figure;
for i = 1:4
    subplot(4, 1, i);
    semilogy(f(2:end), ffts_f{i}(2:end), 'b', 'LineWidth', 1.2);
    hold on;
    semilogy(f(2:end), ffts_h{i}(2:end), 'r--', 'LineWidth', 1.2);
    xlabel('Frequency (Hz)');
    ylabel('Magnitude');
    hold off;
    legend('Faulty', 'Healthy');
    title(sprintf('FFT of signals at %d%% load', volt(i)));
end

```

Code 4.3 Statistical comparison of healthy and faulty vibration signal

```

% Removing DC component
for i = 1:4
    signals_h{i} = signals_h{i} - mean(signals_h{i});
    signals_f{i} = signals_f{i} - mean(signals_f{i});
end

for i = 1:4
    mean_healthy = mean(signals_h{i})
    median_healthy = median(signals_h{i})
    kurt_healthy = kurtosis(signals_h{i})
    skew_healthy = skewness(signals_h{i})

```

```

std_healthy = std(signals_h{i})

mean_misalignment = mean(signals_f{i})
median_misalignment = median(signals_f{i})
kurt_misalignment = kurtosis(signals_f{i})
skew_misalignment = skewness(signals_f{i})
std_misalignment = std(signals_f{i})
end

```

Code 4.4 Calculation and analysis of spectral centroid

```

centroid_f = zeros(1,4); % Allocating space for SC values for faulty signal
centroid_h = zeros(1,4); % Allocating space for SC values for healthy signal

for i = 1:4
    fprintf('Spectral centroid at %d volts\n', i);
    centroid_f(i) = sum(f .* ffts_f{i})/sum(ffts_f{i}) % Calculating spectral centroid
    centroid_h(i) = sum(f .* ffts_h{i})/sum(ffts_h{i})
end

% Plotting SC values in FFT graph
figure;

for i = 1:4
    subplot(4, 1, i);
    f1 = semilogy(f(2:end), ffts_f{i}(2:end), 'b', 'LineWidth', 1.2);
    hold on;
    f2 = semilogy(f(2:end), ffts_h{i}(2:end), 'r-', 'LineWidth', 1.2);

    c1 = xline(centroid_f(i), 'k-', 'LineWidth', 1.2); % Faulty centroid
    c2 = xline(centroid_h(i), 'k-', 'LineWidth', 1.2); % Healthy centroid
    xlabel('Frequency (Hz)');
    ylabel('Magnitude');
    hold off;
    legend([f1, f2, c1, c2], {'0.5mm Fault FFT', 'Healthy FFT', 'Faulty Centroid', 'Healthy Centroid'});
    title(sprintf('FFT of signals at %d%% load and X direction', volt(i)));
    % Here i ran the program three times and switched which axis i had selected in the data
end

```